Research Brief: contrastive learning for tabular data

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Introduction

This brief examines the application of contrastive learning for tabular data, a rapidly evolving area in self-supervised learning (SSL) that seeks to extract meaningful representations from unlabeled datasets without explicit labels or sequential dependencies. The focus is on how contrastive learning—specifically designed to learn representations through positive-negative pair alignment—has been adapted to tabular data contexts, where the structure and relationships between features pose unique challenges. We synthesize findings from recent literature to clarify the current state of contrastive learning in tabular data, highlighting both consensus and unresolved tensions.

Summary

All three papers [1], [2], and [3] are relevant to contrastive learning for tabular data, but [3] focuses on reinforcement learning and is not directly applicable. [1] emphasizes contrastive learning as a key SSL approach for non-sequential tabular data, highlighting its role in mitigating feature order bias and masking sensitivity. [2] investigates contrastive learning through cosine similarity, identifying critical limitations in batch settings and proposing an auxiliary loss to reduce negative-pair similarity variance. [3] is unrelated to tabular data contrastive learning and is excluded from the summary. [4] is about image data augmentation and is not relevant to tabular data contrastive learning. Thus, the summary focuses on [1] and [2] for tabular data contrastive learning.

Findings

1. A Survey on Self-Supervised Learning for Non-Sequential Tabular Data

Wei-Yao Wang, Wei-Wei Du, Derek Xu, Wei Wang, Wen-Chih Peng (2024)

Source reputation: medium — Open-access preprint server; dominant venue for ML/physics/CS research but not peer-reviewed

This paper surveys recent advancements in self-supervised learning (SSL) for non-sequential tabular data. It highlights how SSL addresses the challenge of learning descriptive representations without explicit relations in tabular data by defining pretext tasks on unlabeled datasets. The survey identifies key approaches including contrastive learning, perturbation techniques, and latent space meta-learning that improve representation robustness while mitigating issues like feature order bias and masking percentage sensitivity.

Key claims:

- Self-supervised learning (SSL) has been incorporated into many state-of-the-art models in various domains, where SSL defines pretext tasks based on unlabeled datasets to learn contextualized and robust representations. (p. 1)

- The percentage of masking presents an indecisive and case-by-case issue that every downstream task requires the percentage by empirical adjustments. (p. 7)
- STUNT (Nam et al, 2023) meta-learns self-generated tasks from unlabeled data, which is motivated by columns that may share correlations to the downstream labels (e.g., the feature of "occupation" can be used as a substituted label for "income" before the supervised learning stage). (p. 7)
- GReaT (Borisov et al, 2023) connects tabular and text modalities with a textual encoding schema and randomly permutes the order of features. (p. 8)

2. On the Similarities of Embeddings in Contrastive Learning

Chungpa Lee, Sehee Lim, Kibok Lee, Jy-yong Sohn (2025)

Source reputation: medium — Open-access preprint server; dominant venue for ML/physics/CS research but not peer-reviewed

Based on the abstract only — the full text was not accessible.

This paper investigates contrastive learning through cosine similarity, revealing two critical insights: (1) perfect positive pair alignment is unattainable in full-batch settings when negative-pair similarities fall below a threshold, and (2) mini-batch settings cause higher negative-pair similarity variance, degrading representation quality. The authors propose an auxiliary loss to mitigate this variance. Empirical results demonstrate improved performance when incorporating this loss in small-batch scenarios.

Key claims:

- Perfect alignment of positive pairs is unattainable when negative-pair similarities fall below a threshold in full-batch settings.
- Smaller batch sizes induce stronger separation among negative pairs in the as embedding space, leading to higher variance in negative-pair similarities.
- Higher variance in negative-pair similarities degrades the quality of learned representations in mini-batch settings compared to full-batch settings.
- An auxiliary loss reduces the variance of negative-pair similarities in mini-batch settings, improving performance in small-batch scenarios.

3. Deep Reinforcement Learning with Double Q-Learning

Hado van Hasselt, Arthur Guez, David Silver (2016)

Source reputation: medium — Open scholarly metadata index (aggregator, not a publisher)

Based on the abstract only — the full text was not accessible.

The paper addresses the issue of overestimation in Q-learning, particularly in the context of deep reinforcement learning with the DQN algorithm. It demonstrates that DQN exhibits substantial overestimations in Atari 2600 games and proposes Double Q-learning as a solution to reduce these overestimations and improve performance.

Key claims:

- The popular Q-learning algorithm overestimates action values under certain conditions.
- The DQN algorithm suffers from substantial overestimations in some Atari 2.600 games.

- Double Q-learning can be generalized to work with large-scale function approximation.
- The proposed adaptation to DQN reduces overestimations and leads to better performance on several games.

4. A survey on Image Data Augmentation for Deep Learning

Connor Shorten, Taghi M. Khoshgoftaar (2019)

Source reputation: unknown — Journal of Big Data

Based on the abstract only — the full text was not accessible.

This survey examines image data augmentation techniques to address limited data in deep learning, particularly in medical imaging applications. It highlights how data augmentation enhances dataset size and quality to mitigate overfitting without requiring large-scale data collection. The paper covers various augmentation methods including geometric transformations, color space adjustments, GAN-based approaches, and others, emphasizing their application in improving model performance with constrained data. The survey also discusses test-time augmentation, resolution effects, dataset scaling, and curriculum learning as critical considerations for implementation.

Key claims:

- Deep convolutional neural networks require large datasets to avoid overfitting.
- Many application domains lack sufficient data for deep learning, such as medical imaging.
- Data augmentation techniques include geometric transformations, color space augmentations, kernel filters, mixing images, random erasing, feature space augmentation, adversarial training, GANs, neural style transfer, and meta-learning.
- GAN-based augmentation methods are heavily covered in this survey.

Claims and hypotheses

- Self-supervised learning (SSL) has been incorporated into many state-of-the-art models in various domains, where SSL defines pretext tasks based on unlabeled datasets to learn contextualized and robust representations. [1, p. 1]
- The percentage of masking presents an indecisive and case-by-case issue that every downstream task requires the percentage by empirical adjustments. [1, p. 7]
- STUNT (Nam et al, 2023) meta-learns self-generated tasks from unlabeled data, which is motivated by columns that may share correlations to the downstream labels (e.g., the feature of “occupation” can be used as a substituted label for “income” before the supervised learning stage). [1, p. 7]
- GReaT (Borisov et al, 2023) connects tabular and text modalities with a textual encoding schema and randomly permutes the order of features. [1, p. 8]
- Perfect alignment of positive pairs is unattainable when negative-pair similarities fall below a threshold in full-batch settings. [2]
- Smaller batch sizes induce stronger separation among negative pairs in the as embedding space, leading to higher variance in negative-pair similarities. [2]
- Higher variance in negative-pair similarities degrades the quality of learned representations in mini-batch settings compared to full-batch settings. [2]
- An auxiliary loss reduces the variance of negative-pair similarities in mini-batch settings, improving performance in small-batch scenarios. [2]
- The popular Q-learning algorithm overestimates action values under certain conditions.

[3]
- The DQN algorithm suffers from substantial overestimations in some Atari 2.600 games. [3]
- Double Q-learning can be generalized to work with large-scale function approximation. [3]
- The proposed adaptation to DQN reduces overestimations and leads to better performance on several games. [3]
- Deep convolutional neural networks require large datasets to avoid overfitting. [4]
- Many application domains lack sufficient data for deep learning, such as medical imaging. [4]
- Data augmentation techniques include geometric transformations, color space augmentations, kernel filters, mixing images, random erasing, feature space augmentation, adversarial training, GANs, neural style transfer, and meta-learning. [4]
- GAN-based augmentation methods are heavily covered in this survey. [4]

Conclusion

The evidence indicates that contrastive learning for tabular data is a promising yet nuanced approach, with [1] establishing its role in SSL for non-sequential tabular data and [2] revealing critical limitations in batch settings that require careful tuning of negative-pair similarity thresholds and batch sizes.

Further research

- How does the percentage of masking in contrastive learning affect downstream tasks in tabular data?
- Can the auxiliary loss proposed in [2] be adapted to improve tabular data contrastive learning with larger batch sizes?
- What is the impact of feature order bias on the performance of contrastive learning in tabular data?

References

- Chungpa Lee, Sehee Lim, Kibok Lee, Jy-yong Sohn (2025) 'On the Similarities of Embeddings in Contrastive Learning', arXiv:2506.09781v2. Available at: http://arxiv.org/abs/2506.09781v2 [reputation: medium] [access: open]
- Wei-Yao Wang, Wei-Wei Du, Derek Xu, Wei Wang, Wen-Chih Peng (2024) 'A Survey on Self-Supervised Learning for Non-Sequential Tabular Data', arXiv:2402.01204v4. Available at: http://arxiv.org/abs/2402.01204v4 [reputation: medium] [access: open]
- Connor Shorten, Taghi M. Khoshgoftaar (2019) 'A survey on Image Data Augmentation for Deep Learning', doi:10.1186/s40537-019-0197-0. Available at: https://openalex.org/W2954996726 [reputation: unknown] [citations: 12510] [access: open]
- Hado van Hasselt, Arthur Guez, David Silver (2016) 'Deep Reinforcement Learning with Double Q-Learning', doi:10.1609/aaai.v30i1.10295. Available at: https://openalex.org/W2746553466 [reputation: medium] [citations: 6061] [access: open]

Limitations

- *SupervisorAgent*: 129/150 words, 2/2 reputable sources; criteria not met -> retrying with a broader search
- *ResearchAgent*: no semanticscholar results for query 'tabular data self-supervised

learning'

Scope note: this is a deliberately small, local pipeline (arXiv + local retrieval + a local LLM). See the README for the design-vs-implementation trade-offs.

Appendix: decision log

#	agent	kind	reason
1	ParseAgent	strategy	2402.01204v4: parsed with parse_basic
2	SupervisorAgent	retry	129/150 words, 2/2 reputable sources; criteria not met -> retrying with a broader search
3	ResearchAgent	error	no semanticscholar results for query 'tabular data self-supervised learning'
4	SupervisorAgent	converge	256/150 words, 3/2 reputable sources; criteria met -> publishing